Earth's New Mini-Moon Is Staying for Christmas

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Earth has a new temporary neighbor, and it is staying for the holidays. Meet 2026 QQ8, a space rock only about 6 feet across. Scientists first spotted it in August 2026, after it slipped into orbit around Earth in April 2026. The International Astronomical Union's **Minor Planet Center** reported the discovery on October 29. QQ8 is called a **mini-moon**, or a **temporary satellite**, because Earth's gravity has caught it for a short visit. It will stay through December, then drift away in 2027. Scientists have worked out its path around the sun, which takes 359 days, or about 0.98 of an Earth year. The rock swings close to Earth about once every ten years, but this time it came close enough for Earth's gravity to grab hold. At its nearest, QQ8 will pass about 353,000 miles from our planet. That is about 110,000 miles farther away than the average distance between Earth and the moon. There is no need to worry about a crash. Even if QQ8 headed straight for Earth, it would likely burn up completely in the **atmosphere** before reaching the ground. QQ8 is not the first **mini-moon**. Research last year suggested Earth could have as many as six **mini-moons** at any given time, and many of them may be **fragments** of our own moon. QQ8 is bigger than most. Modelling earlier this year suggested the average **mini-moon** is only about 75 centimeters across, or 30 inches, and the average stay lasts around nine months. Captured in April 2026, it should move on in January 2027. The same research suggested that a rock about 10 feet wide could stay caught for 10 to 20 years. For now, though, QQ8 is a short-term guest, a tiny traveler spending Christmas with Earth before heading back out into space.

Word Treasure

mini-moon -- A small space rock that Earth's gravity catches for a short visit.

temporary satellite -- An object that orbits a planet for a while instead of staying there forever.

Minor Planet Center -- The group that records the discovery of small objects in space.

fragments -- Small broken pieces of a larger object.

atmosphere -- The layer of gases that surrounds Earth.

Background you need

A mini-moon is a small space rock that Earth's gravity catches for a short time, so it circles Earth like a tiny temporary moon instead of just flying past.

This happens when a rock traveling around the Sun comes close enough for Earth's pull to hold it for a while, even though it is not trapped forever.

Most of these visitors are very small and stay only for a short visit before Earth's gravity lets go and they drift back out into space.

Scientists study them because they can show how small rocks move near Earth and where some of them may have come from.

Break it down

LEAD: Earth has a temporary mini-moon, 2026 QQ8, that will stay through December before drifting away in 2027.

KEY EVENT: Earth's gravity captured the space rock in April 2026.

+- WHO: 2026 QQ8, a space rock about 6 feet across

+- WHEN: Spotted in August 2026 and reported on October 29 by the International Astronomical Union's Minor Planet Center

+- HOW LONG: It is expected to move on in January 2027

+- HOW CLOSE: At its nearest, it will pass about 353,000 miles from Earth

EVIDENCE: Scientists worked out its 359-day path around the sun, and earlier modelling put the average mini-moon at about 75 centimeters across with an average stay of around nine months.

WHAT HAPPENS NEXT: QQ8 will drift away in 2027, while a rock about 10 feet wide could stay caught for 10 to 20 years.

Why it matters

Mini-moons show how Earth's gravity can briefly capture passing space rocks, and studying them helps scientists understand near-Earth objects and where some of them may have come from.

Test what you remember

1. **When did scientists first spot 2026 QQ8?**
   - (A) In April 2026, when it entered orbit
   - (B) In August 2026
   - (C) In January 2027, when it moves on
   - (D) On October 29, when its discovery was reported
2. **How wide is 2026 QQ8?**
   - (A) About 75 centimeters
   - (B) About 6 feet
   - (C) About 30 inches
   - (D) About 10 feet
3. **How long does 2026 QQ8 take to travel around the sun?**
   - (A) About nine months
   - (B) 359 days
   - (C) 10 to 20 years
   - (D) About ten years
4. **At its nearest, about how far will QQ8 pass from Earth?**
   - (A) About 353,000 miles
   - (B) Farther than the moon, but no distance is given
   - (C) Close enough to reach the ground
   - (D) About 110,000 miles
5. **What would most likely happen if QQ8 headed straight for Earth?**
   - (A) It would crash into Earth
   - (B) It would become a permanent moon
   - (C) It would burn up completely in the atmosphere before reaching the ground
   - (D) It would stay in orbit for 10 to 20 years
6. **According to the research described, how long could a rock about 10 feet wide stay caught by Earth's gravity?**
   - (A) Through December only
   - (B) About 359 days
   - (C) About nine months
   - (D) 10 to 20 years

Answer key (try the quiz first!): 1: B 2: B 3: B 4: A 5: C 6: D

Questions to think about

MY THOUGHTS (at least 20 words):